

ZS-A28

The Absolute Vacuum-Energy Scale B3: Why Microscopic Scale-Generation Fails, and the X–Y Holographic Boundary Law

Kenny Kang · Z-Spin Cosmology Collaboration · June 2026 · Theme: Astrophysics · **ZS-A28 v1.7**
Verification: 94/94 PASS (fail-closed, real asserts) | **Zero Free Parameters** | ($A = 35/437$, $Q = 11$, $\dim Z = 2$)
LOCKED | verify_zs_a28_v1_7.py SHA256: 6278e291...e40c7d

*Status of the central claim. The absolute vacuum-energy scale **B3** remains **OPEN** (forbidden in the absolute sense by the A–Q-Only No-Go), but is transformed: Part I shows every microscopic engine merely re-expresses one number N ; Part II reframes B3 as a UV/IR boundary, dissolving the hierarchy into $\rho_\Lambda'^4 = \sqrt{(\tilde{M}_P H_0)}$. **v1.6 grounds 83 in its corpus derivation** (ZS-F2→U4→A19→A20; $83 = 82+1$), upgrades the assignment direction to **DERIVED**, retires XB-3 as inapplicable via the constant- Λ de Sitter entropy (resolving the ZS-U4 conflict; $c^2=83/121$ is $\Omega_{\Lambda,0}$ today, not the de Sitter coeff 1), makes Theorem A28.10 a conditional implication, and corrects the ΔN_{eff} statement: $\Delta N_{\text{eff}}^{\text{BBN}} = 0.160$ survives (D/H) while always-on $\Delta N_{\text{eff}}^{\text{CMB}} = 0.160$ is **FALSIFIED** ($\Delta\chi^2 = +408$); the clean channels are $\Delta N_{\text{eff}}^{\text{BBN}}$ and $r = 0.0089$.*

Corrections, resolution, unification, reframe, and repair of record (v1.0 → v1.7)

GROUNDING (v1.6). **83 is DERIVED in the corpus**, not introduced by A28: ZS-F2 §11.4 face-counting ($\Omega_b = XZ/Q^2 = 6/121$, $\Omega_{\text{cdm}} = F(\text{truncated icosahedron})/Q^2 = 32/121$, cross-check $XQ-1 = 32$) → ZS-U4 (locked late-time budget) → ZS-A19 (rank-32 conserved dust) → ZS-A20 ($\rho_b/\rho_c = 6/32$ at action level). The dark-energy entry **83 = 121 – 6 – 32 = 82 + 1 = $Q^2 - X(Z+Q) + \text{ind.}(D_Z)$** : a trace *complement*, and the assignment direction is theorem-forced (baryon = X-sector current → 6; CDM = unique I_h mediator → 32), so $c^2 = 83/121$ is **DERIVED-COND** — not a hidden fit. **XB-3 retired-as-inapplicable**: corpus DE is constant Λ ($w = -1$, ZS-U1/U4) and the holographic scaling is static (de Sitter entropy, A26/A27) — no Hubble-vs-event-horizon choice; the dynamical $w(z)$ detour is dropped, resolving the U4 conflict. ΔN_{eff} corrected: $\Delta N_{\text{eff}}^{\text{BBN}} = 0.160$ survives (D/H), but always-on $\Delta N_{\text{eff}}^{\text{CMB}} = 0.160$ is **FALSIFIED** ($\Delta\chi^2 = +408$); the clean channels are $\Delta N_{\text{eff}}^{\text{BBN}}$ and $r = 0.0089$. Theorem A28.10 is now a *conditional implication*, not an iff. §0, §12–§15.

REPAIRED (v1.5). The v1.4 unification Theorem A28.10 *conflated* two different objects: $c^2 = \Omega_\Lambda = 83/121$ is a **$Q^2 = 121$ face budget** (the 9-block tensor $(3,2,6) \otimes (3,2,6)$, with Λ the remainder 83), whereas Condition C is the **$Q = 11$ sector weight** $(3,2,6)/11$ — and no sector weight equals $83/121$. v1.5 *decouples* them: c^2 needs only the face budget, and Condition C becomes a *separate* (still open) sector-weight problem — a genuine simplification. Also: $(3,2,6)/11$ (Hilbert register) and $(3,3,5)/11$ (operator algebra A_{ZS}) are distinct objects (A19.NG1 resolved); and the corpus CDM is *conserved dust* (A19, no coupling), so the reframe's $w(z)$ is vanilla and DESI-disfavored — registered honestly. §12–§14.

REFRAMED (v1.4). Part I (the microscopic route) is preserved as an audit; Part II reframes B3 as a UV/IR boundary problem — the hierarchy $N = 69.16 = \frac{1}{2} \ln(\tilde{M}_P/H_0)$ is a ratio of scales, so the tuning collapses to one $O(1)$ number, $c^2 = 83/121$.

UNIFIED (v1.3) / **RESOLVED** (v1.2) / **RETIRED**: both internal engines re-express N (engine swap relabels, not closes); the chiral-Pfaffian $\frac{1}{2}$ is not forced; the spectral route is retired.

RETRACTED (v1.0 §7): non-unitary torsion. **DEMOTED**: v1.0 Thms A28.2/A28.4/A28.3/A28.5; B3-4 PARTIAL.

PRESERVED: Thm A28.1 (mass-dimension correction) **PROVEN**.

§0. Abstract

ZS-A27 reduced the absolute vacuum-energy scale B3 to a four-root programme; ZS-A28 v1.0–v1.3 attacked the microscopic sequestering trunk and proved, blind, that the internal scale-generation route is **closed-negative** — the chiral Pfaffian $\frac{1}{2}$ is fitting, not forced, and swapping the two internal engines merely relabels one number $N = \ln(\bar{\mathbf{M}}_p/\rho_\Lambda^{1/4}) \approx 69.16$ without closing it. Part I is preserved as that audit.

Part II reframes the problem: vacuum energy is a UV/IR boundary quantity, not a microscopic constant. $N = \frac{1}{2} \ln(\bar{\mathbf{M}}_p/H_0) - \frac{1}{4} \ln(3c^2)$, so $\rho_\Lambda^{1/4} = \sqrt[4]{(\bar{\mathbf{M}}_p H_0)}$ is a geometric mean and the ~ 120 -order tuning collapses to one $O(1)$ number $c^2 = \Omega_\Lambda$.

83 is DERIVED in the corpus, not introduced here. ZS-F2 §11.4 face-counting gives $\Omega_b = XZ/Q^2 = 6/121$ (baryon = X-sector matter with a Z_2 seam charge, **DERIVED**) and $\Omega_{\text{cdm}} = F(\text{truncated icosahedron})/Q^2 = 32/121$ (CDM = the *unique* I_h mediator; cross-checked $XQ-1 = 32$); ZS-U4 adopts the budget $(6,32,83)/121$ as the locked late-time input; ZS-A19/A20 realise 32 as a conserved rank-32 boundary-tension dust and derive $\rho_b/\rho_c = 6/32 = 3/16$ at action level. The dark-energy entry **83 = 121 – 6 – 32** is therefore the trace *complement* of the 38 matter modes, with the deeper Z-Spin identity **83 = Q² – X(Z+Q) + ind.(D_Z) = 82 + 1** (82 = pre-gauge non-matter complement; +1 = the Z_2 -odd mediation mode removed from CDM, 33→32) — a re-expression of the same complement, *not* independent forcing (which lies entirely in 6 and 32). The assignment direction (visible = 6, dark = 32) is *theorem-forced* (CDM = 32 a polyhedral theorem; baryon = 6 the softer ZS-F5 B3.1 leg), so $c^2 = 83/121$ is **DERIVED-COND** — not a hidden fit. The genuine residuals are the equal-weight embedding Φ_{face} (categorical state-sum polish) and the **Vacuum Identification** $q_\Lambda \rightarrow T_{\mu\nu}^\Lambda = -\rho_\Lambda g_{\mu\nu}$, which is **OPEN**. ($83/38 = 2.184$ is a 0.8% near-match to $2e^\Lambda = 2.167$, not an identity — 83 is not explained via $2e^\Lambda$.)

XB-3 is retired-as-inapplicable. The corpus dark energy is constant Λ ($w = -1$ exactly, ZS-U1/U4); the holographic scaling $\rho_\Lambda \propto \bar{\mathbf{M}}_p^2 H^2$ arises *statically* from the de Sitter horizon entropy (the A26/A27 dilation-anomaly capstone, $\rho_\Lambda/\bar{\mathbf{M}}_p^4 \approx 24\pi^2/S_{\text{ds}}$), where the Hubble, apparent, and event horizons coincide — so there is no cutoff to choose (constant Λ removes the dynamical-cutoff problem; it does not derive a horizon). Here $c^2 = 83/121$ is TODAY's fraction $\Omega_{\Lambda,0}$; the de Sitter saturation coefficient is 1 with $H_\infty = \sqrt[4]{(83/121)} H_0$ — distinct objects. The dynamical-holographic $w(z)$ detour of earlier drafts is dropped, which *resolves* the version conflict with ZS-U4. The unification **B3** $\Leftarrow$ [**Frame H** + Φ_{face} + **horizon-source coupling**] is a *conditional implication, not an iff* ($c^2 > 0$ alone does not manufacture an action-level Λ source). B3-relational [$\rho_\Lambda = 3c^2 \bar{\mathbf{M}}_p^2 H^2$] is **DERIVED-COND**; B3-absolute [predicting H_0 from **A**, **Q**] remains a **No-Go**.

Predictions. The clean falsifiable channels are $\Delta N_{\text{eff}}^{\text{BBN}} = \dim(\mathbf{Z}) \cdot \mathbf{A} = 2\mathbf{A} = 0.160$ (resolving the D/H tension to -0.05σ) and $r = 0.0089$ (tensor-to-scalar; high-significance at LiteBIRD nominal sensitivity, $\delta r < 0.001$). The *always-on* $\Delta N_{\text{eff}}^{\text{CMB}} = 0.160$ scenario is **FALSIFIED** by the corpus Cobaya analysis ($\Delta\chi^2_{\text{CMB}} = +408$); CMB-S4's role is to distinguish the BBN-vs-CMB activation history, not to 5σ -detect 0.160. All results are verified fail-closed (**94/94** asserts, after retiring v1.6's contradictory dynamical- $w(z)$ checks). As a capstone, A28 classifies B3 four ways — B3-A (A–Q-only absolute) **CLOSED-NEGATIVE**; B3-B (with an extra dimensionful datum) **OPEN**; B3-C (today's relation) **DERIVED-COND**; B3-D (the vacuum-identification bridge = Premise P0) **OPEN** — so B3 is *not closed but reframed, grounded, and localized*. (**A**, **Q**, $\dim \mathbf{Z}$) = (35/437, 11, 2) **LOCKED**.

Epistemic Status Legend

PROVEN — established by internal computation/algebra at machine precision.

DERIVED-CONDITIONAL — DERIVED given explicitly named external/corpus conditions.
HYPOTHESIS — well-motivated, pre-registered, not yet derived; falsifiable.
PARTIAL — a closure path is specified and partly executed; the load-bearing computation is pending.
TARGET-COMPUTATION — a finite, decidable computation set up but deferred (here, to lattice / ZS-A29).
OPEN — not closed by current corpus tools; no committed probability.
RESOLVED — a v1.1 open question now decided here (here: closed-negative for the internal route).
REPAIRED — a v1.4 claim corrected here (the face-budget vs sector-weight conflation in Theorem A28.10).
REFRAMED / RETIRED / RETAINED / DECOUPLED — a Part-I gate recast under the Part-II reframe (recast / dropped from the scale problem / still needed / separated).
RETRACTED / DEMOTED — a v1.0 claim withdrawn or lowered in status by audit.

§1. Introduction: from three weak handles to physical objects

ZS-A27 §7 proved the **A–Q-Only Scale-Generation No-Go**: the locked dimensionless data cannot fix a dimensionful ρ_Λ . This is not a wall — it says only that A and Q alone are insufficient; closing B3 with the one locked dimensionful anchor $\bar{\mathbf{M}}_p$ plus a dynamical scale-generating mechanism does not violate it. ZS-A28 v1.0 attacked the sequestering trunk in the order ZS-A27 §8.2 prescribed and found a real dimensional error in A27 (the source term $\bar{\mathbf{M}}_p^2 \rightarrow \bar{\mathbf{M}}_p^{-2}$). But an external audit showed v1.0's trunk rested on three weak handles: a Wilson-LOCATOR Gram matrix standing in for the kernel, a *declared* membrane branch standing in for branch selection, and a torsion route mis-labelled ‘non-unitary’. v1.1 retracts these (box above) and rebuilds on physical objects.

The reorganisation is driven by one observation: the abstract odd operators Ω_a of v1.0 have two canonical, physical realizations in four dimensions — the **gauge** and **gravitational Pontryagin densities**,

$$\Omega_1^- = (1/32\pi^2) \operatorname{tr} F_- \wedge F_-, \quad \Omega_2^- = (1/384\pi^2) R \wedge R, \quad (\text{A28.30})$$

both pseudoscalar (J_Z -odd, since $\gamma^5 = J_Z$ and the chiral anomaly is $\partial_\mu j_5^\mu \propto \operatorname{tr} F \tilde{F}$), gauge/diffeomorphism invariant, dimension 4, and expressible as four-forms with well-defined *topological susceptibility*. This single substitution makes B3-2, B3-4, and B3-5 the same physics, and is what the rest of the paper develops.

Table 1.1. v1.0 weak handle $\rightarrow$ v1.1 physical object, with the new status.

v1.0 handle	v1.1 physical object	Gate	Status
abstract Ω_a (arithmetic legs)	Pontryagin densities $\operatorname{tr} F \tilde{F}$, $R \tilde{R}$	B3-2	DERIVED-COND
Wilson-LOCATOR Gram K_-	susceptibility Hessian χ_{ab}/Λ_-^4	B3-2/5	DERIVED-COND
‘dim $Z=2$ so $SU(2)$ ’	symplectic selection from Kramers $C^2=-1$	B3-4	DERIVED-COND
declared membrane branch	equilibrium q-boundary (susceptibility path)	B3-3	PARTIAL
non-unitary CM torsion	chiral Pfaffian $\frac{1}{2}$ of J_Z -graded det	B3-5	HYPOTHESIS

§2. Preserved: the corrected doublet action (PROVEN)

The one v1.0 result that survives audit unchanged is the dimensional correction of record. With $[q_a] = 2$, $[\mathcal{Q}_a] = 4$, the dimensionally consistent J_Z -odd action and its variation are

$$S_- = \int d^4x \sqrt{-g} \left[-\frac{1}{2} q_a (K_q^{-1})_{ab} q_b + \bar{\mathbf{M}}_P^{-2} q_a \mathcal{Q}_a \right], \quad q_a = \bar{\mathbf{M}}_P^{-2} (K_q)_{ab} \mathcal{Q}_b, \quad (\text{A28.1-2})$$

$$\rho_{M1} = (2 \bar{\mathbf{M}}_P^4)^{-1} \mathcal{Q}^T K_q \mathcal{Q} \geq 0 \Leftrightarrow K_q \geq 0. \quad (\text{A28.3})$$

Positivity requires a PSD kernel; v1.0 supplied it as a Gram ansatz, v1.1 supplies it *by construction* (§4). With $\chi_Z = -1$, a single four-form gives $\rho < 0$, so the doublet structure is mandatory (Corollary, **PROVEN**).

§3. Root R0: the sequestering trunk, honestly bounded (PARTIAL)

v1.0 claimed the Selective Sequestering Compatibility Theorem at the action level. The audit is correct that this was over-stated: the unified action was a component list, and the survival of the odd residual was *coded as an identity* rather than *derived* by metric variation — and $q = *F_4$ has a metric-dependent Hodge dual that a complete stress tensor must include. v1.1 states the correct, bounded claim and the explicit computation that would close it.

The mechanism is sound and externally anchored: Kaloper–Padilla (and the graviton-loop Omnia-Sequestra extension) [3,4] remove *every* constant vacuum shift, leaving an *a priori arbitrary* finite radiatively-stable residual; Klinkhamer–Volovik q-theory [5–7] performs the equilibrium nullification (the corpus $\rho_Z = 0$, ZS-F9, **PROVEN**) and the electroweak ‘kick’ $\Lambda \sim E_{\text{ew}}^8 / \bar{\mathbf{M}}_P^4$ [6]. The selectivity is by J_Z grading: the even part self-nullifies, the odd part survives.

What must be computed to close B3-1 (deferred to ZS-A29). Write the Omnia-Sequestra action in full and vary $g_{\mu\nu}$, Λ , κ^2 , θ_+ , A_3^+ , $A_{3,a}^-$ to obtain

$$\bar{\mathbf{M}}_P^2 G_{\mu\nu} = T_{\mu\nu} - \frac{1}{4} \langle T \rangle g_{\mu\nu} - \Delta \Lambda_{\text{GB}} g_{\mu\nu} - E_-(\theta^*) g_{\mu\nu}, \quad (\text{A28.31})$$

with $E_-(\theta^*)$ the odd-sector topological energy arising *not* from the matter stress tensor but from the separate global four-form sector — exactly so that the historic average $\langle T \rangle$ removes the even constants but cannot reach E_- . The closure conditions are: (i) $L_m \rightarrow L_m + \Delta V$ leaves (A28.31) invariant; (ii) the odd topological energy is not removed; (iii) no independent additive constant survives in the odd sector; (iv) matter *and* graviton counterterms are sequestered. Until (i)–(iv) are checked by symbolic variation and an ADM/Hamiltonian-constraint analysis, R0 is **PARTIAL**: a specified path, not a theorem. (This is the honest replacement for v1.0’s coded identity.)

§4. Root M1-kernel: the topological-susceptibility Hessian (Theorem A28.8)

v1.0 computed the rank of a Wilson-LOCATOR Gram matrix and called it the vacuum-action Hessian. v1.1 uses the physically correct object. Define the topological charges and their susceptibility matrix

$$\mathcal{Q}_a = \int_{M^4} \mathcal{Q}_a^-(x) d^4x, \quad \chi_{ab} = V_4^{-1} \langle \mathcal{Q}_a \mathcal{Q}_b \rangle_c = \partial^2 E(\theta) / \partial \theta_a \partial \theta_b |_{\theta=0}, \quad (\text{A28.32})$$

the second θ -derivative of the vacuum energy. The dimensionless kernel is $K_{ab}^{\text{vac}} = \chi_{ab} / \Lambda_-^4$.

Theorem A28.8 (Susceptibility Kernel, DERIVED-CONDITIONAL). (a) χ_{ab} is a covariance matrix of the real charges $\mathcal{Q}_a$, hence **positive-semidefinite by construction**:

$$c^T \chi c = V_4^{-1} \langle (\sum_a c_a \mathcal{Q}_a)^2 \rangle_c \geq 0 \quad \text{for all } c. \quad (\text{A28.33})$$

(b) Basis covariance is automatic: $\theta \rightarrow U\theta \Rightarrow \chi \rightarrow U\chi U^\dagger$. (c) $\text{rank } \chi \geq 1$ *if and only if* $\chi_{11} = V_4^{-1} \langle \mathcal{Q}_{\text{YM}}^2 \rangle_c > 0$ — the Yang–Mills topological susceptibility, which is *positive* (a robust lattice result for pure gauge

theory). rank 2 iff $\det \chi = \chi_{11}\chi_{22} - |\chi_{12}|^2 > 0$. **DERIVED-CONDITIONAL** on $\chi_{11} > 0$ (external) and on the actual χ_{ab} being a **TARGET-COMPUTATION** (pure-SU(2) lattice).

The unification (the structural gain). This χ is not an auxiliary kernel: it is the *same* quantity that fixes the residual coefficient. The residual scaling is $\rho_\Lambda = C_- \Lambda_-^8 / \bar{\mathbf{M}}_p^4$ with C_- determined by χ_{ab} and the q-crossover dynamics. Therefore **B3-2 (rank ≥ 1) and B3-5 (coefficient) are one physical quantity** — they close, or fail, together. This is the replacement for v1.0's two unrelated handles, and it is why the LOCATOR Gram matrix is removed from the proof entirely (arithmetic conductors now enter only in the interpretation of §7).

§5. Root B3-4: Symplectic Selection of SU(2)_Z (Theorem A28.7)

The hidden odd gauge factor G_- was fully open in v1.0. v1.1 *forces* it, and ties it to the foundational Z-spin. The argument is short and rigorous.

The odd cohomology is two-complex-dimensional, $H_D^- \cong \mathbb{C}^2$ (the Kostant realization, ZS-M27). The Z-sector carries a **spin-1/2** structure ($\chi_Z = -1$ is the half-angle lift, ZS-Q14 Lemma 5.1). A spin-1/2 system has a **Kramers** time-reversal / charge-conjugation operator C that is antilinear with

$$C^2 = -1 \quad (\text{Kramers degeneracy, the quaternionic real structure of half-integer spin}). \quad (\text{A28.34})$$

Theorem A28.7 (Symplectic Selection, DERIVED-CONDITIONAL). Let h be the Hermitian form on $H_D^- \cong \mathbb{C}^2$ and C the Kramers structure (A28.34); set the symplectic form $\omega(u, v) = h(Cu, v)$. The connected group of $\mathbb{C}$ -linear maps preserving both h and ω is

$$\text{Aut}(h, \omega) = \text{USp}(2) \cong \mathbf{SU}(2)_Z. \quad (\text{A28.35})$$

Proof (verified symbolically). Work in a basis with $h = I$ and $C = J_0 K$, $J_0 = [[0, 1], [-1, 0]]$, K complex conjugation, so $J_0^2 = -I$ gives $C^2 = -I$. The Lie algebra of h -preserving maps is the anti-Hermitian matrices; imposing commutation with the antilinear C ($X J_0 = J_0 \bar{X}$) forces X traceless. The solution space is the traceless anti-Hermitian matrices — exactly $su(2)$, three-dimensional, spanned by $i\sigma_x, i\sigma_y, i\sigma_z$. (verify `zs_a28_v1_1.py`, T1.)

Why this is stronger than ‘dim Z = 2 $\Rightarrow$ SU(2)’. Three independent inputs are used: the *irreducible* $\mathbb{C}^2$ (ZS-M27), the $C^2 = -1$ quaternionic structure (ZS-Q14 spin-1/2), and *unitary/symplectic* preservation. Together they single out $\mathbf{SU}(2)_Z$ — the ‘Z’ in the gauge group is literally the Z-spin. The load-bearing condition is (A28.34); it is **DERIVED-CONDITIONAL** on identifying the odd-sector C with the ZS-Q14 Kramers operator.

Immediate consequences (PROVEN given SU(2)). Pure SU(2) Yang–Mills has $b_0 = (11/3)C_2(\text{adj}) = 22/3 > 0$ (asymptotic freedom is automatic, no fitting), $b_1 = 136/3 > 0$, and — being *pure bosonic* — carries *no fermion doublets*, so the Witten SU(2) global anomaly (which requires an even doublet count) is trivially absent. This is the right matter content for B3-4 and simplifies B3-6.

The coupling (v1.3): B3-4 demands not only G_- but $g_-(\mu_0) = f(\mathbf{A}, \mathbf{Q})$. Under the common spectral normalization $1/g_i^2(\mu_0) = N_{\text{norm}} I_i$ with I_i the Dynkin index, and with both the visible $\text{SU}(2)_L$ and the hidden $\text{SU}(2)_Z$ acting on a *fundamental doublet* ($I = 1/2$), the common N_{norm} cancels in the ratio and

$$g_Z^2(\mu_0)/g_2^2(\mu_0) = I_2/I_Z = 1 \Rightarrow g_Z(\mu_0) = g_2(\mu_0). \quad (\text{A28.40})$$

This is a *falsifiable prediction*, not a free parameter: the hidden coupling equals the Standard-Model $\text{SU}(2)_L$ coupling at the matching scale. Running the latter to $\bar{\mathbf{M}}_p$ (one-loop, $b_2 = 19/6$) gives $g_Z^2(\bar{\mathbf{M}}_p) \approx 0.258$. **But the consequence is decisive:** with $b_0 = 22/3$ and this coupling, dimensional transmutation

gives a confinement scale $\Lambda_Z = \bar{\mathbf{M}}_p \exp[-8\pi^2/(b_0 g_Z^2)] \approx \mathbf{1.9 \text{ GeV}}$ — *not* the $\sim 2.3 \text{ TeV}$ crossover the observed ρ_Λ requires, and a residual $\rho_\Lambda^{1/4} \approx 1.5 \times 10^{-9} \text{ eV}$ that undershoots observation by $\sim 10^6$. So $\text{SU}(2)_Z$ *confinement* does not set the dark-energy scale (channel (i) is dead at the natural coupling); B3-4 is **PARTIAL** — the group and coupling are fixed, but the bundle, the kinetic term, and the scale consequence are not closed.

§6. Roots B3-3 and B3-6: the susceptibility path and the honest anomaly status

B3-3 (re-opened, with a stronger path). v1.0 declared a membrane branch; quantization ($n \in \mathbb{Z}$) does not select one. v1.1 takes the *susceptibility* path instead of the flux-branch path: exact J_Z fixes $\langle \Omega_- \rangle = 0$ (the odd one-point vanishes), while the residual lives in the *nonzero susceptibility* $\chi_- = \langle \Omega_- \Omega_- \rangle_c > 0$ and a nonequilibrium kick. The proposed closure (B3-3') is: J_Z symmetry, regularity, and a high-temperature equilibrium initial condition uniquely fix all q / four-form integration constants, and the late residual is the *unique solution* of the q -equation of motion through the crossover,

$$\ddot{q} + 3H\dot{q} + \partial \varepsilon(q, T) / \partial q = 0, \quad q(t \ll t_c) = q_{\text{eq}}(T), \quad \dot{q} = 0, \quad (\text{A28.36})$$

not an arbitrary branch. This removes the discrete free parameter *only if* the equilibrium initial condition is shown to fix $\theta_- = 0$ uniquely — a **PARTIAL** result (the q -dynamics $\varepsilon(q, T)$ and the coefficient must be derived, not imported). It is, however, more physical than a declared branch: Klinkhamer–Volovik show the crossover *does* leave a residual $\sim E_{\text{cross}}^8 / \bar{\mathbf{M}}_p^4$ [6]; Z-Spin must supply its coefficient and dynamics.

B3-6 (honest status). v1.0's parity table is correct as far as it goes — every degree-one-in-odd mixing ($q_+, q_-, q_R, q_{\hat{H}H}$) is J_Z -odd and forbidden — but a parity table is *not* a one-loop proof, and $\det(J_Z|_{\text{odd}}) = +1$ is *not* a proof of anomaly freedom. The correct closure needs (i) an algebraic-renormalization computation of the BRST cohomologies $H^0(s|d)$ (allowed counterterms) and $H^1(s|d)$ (local anomaly), to operator dimension 8; (ii) an all-order Ward identity $\Gamma[\Phi_+, \Phi_-] = \Gamma[\Phi_+, -\Phi_-]$; and (iii) a **Dai–Freed η -invariant** check $\exp(2\pi i \eta_{\text{SD}}[J_Z, G_-]) = 1$ over the spin-bordism generators. The *pure bosonic* $\text{SU}(2)_Z$ of §5 makes (iii) tractable. Status: **OPEN** with a definite, simplified program (a correction of v1.0's **DEMOTED** claim).

§7. Root S1: the scale reduces to one number, and the chiral Pfaffian $\frac{1}{2}$ (Theorem A28.9)

With the torsion mis-label retracted, the scale route is dimensional transmutation of the $\text{SU}(2)_Z$ factor: $\Lambda_- = \bar{\mathbf{M}}_p \exp[-8\pi^2/b_0 g_-^2]$, with $b_0 = 22/3$ (§5) and g_- from the spectral-action trace. The auditor's arithmetic (reproduced fail-closed) shows the *entire* terminal scale reduces to a single spectral log-determinant $C_{\text{odd}}^{\text{sp}}$:

$$E^* = \bar{\mathbf{M}}_p \exp(-\gamma_{\text{CW}} C^{\text{sp}}), \quad \rho_\Lambda^{1/4} = E^*/\bar{\mathbf{M}}_p = \bar{\mathbf{M}}_p \exp(-2\gamma_{\text{CW}} C^{\text{sp}}), \quad \gamma_{\text{CW}} = 38/9. \quad (\text{A28.37})$$

The *even* invariant $C_{\text{even}}^{\text{sp}} = 11 \ln 2 + \ln 3 = 8.723$ reproduces the electroweak vev $E^* \approx 246 \text{ GeV}$ (this anchors $\gamma_{\text{CW}} = 38/9$), but gives $\rho_\Lambda^{1/4} \approx 2.48 \times 10^{-5} \text{ eV}$ — **90× short**. The gap closes if $\ln 3$ is *halved*:

$$C_{\text{odd}}^{\text{sp}} = 11 \ln 2 + \frac{1}{2} \ln 3 = 8.174 \Rightarrow \rho_\Lambda^{1/4} = 2.57 \times 10^{-3} \text{ eV} = 1.15 \times \text{observed} \quad (0.06 \text{ decade}). \quad (\text{A28.38})$$

Theorem A28.9 (Chiral Pfaffian $\frac{1}{2}$, structure, PROVEN; assignment, HYPOTHESIS). The $J_Z = \gamma^5$ -graded (chiral) determinant is a **Pfaffian**: for the antisymmetric operator of a chiral complex, $\text{Pf}^2 = \det$, so

$$\log \text{Pf} = \frac{1}{2} \log \det, \quad (\text{A28.39})$$

a structural factor $\frac{1}{2}$ that lives in the chiral grading (*not* in any non-unitarity — that was the v1.0 error). This is the legitimate origin of a $\frac{1}{2}$ in $C_{\text{odd}}^{\text{sp}}$, and ZS-A27 §7.1 already notes a Pfaffian $\frac{1}{2}$ is *not* numerology *provided* it falls out of the measure a priori.

The decisive computation (v1.2), executed blind. v1.1 left one question: does the J_Z -graded measure *force* the $\frac{1}{2}$ onto $\ln 3$ (the conductor-3 leg) rather than $\ln 2$ (the $Q = 11$ register)? We enumerate the natural chiral/reality-type assignments and compute $\rho_\Lambda^{1/4}$ for each — both-full = $8.723 (\times 0.011)$, global-Pfaffian = $4.36 (\times 10^{14})$, $\ln 3$ -halved = $8.174 (\times 1.15)$, register-halved = $4.91 (\times 10^{12})$. **Exactly one** assignment matches, which is suggestive — but it is *fitting* unless an independent object selects it. The reality structure of the doublet does *not* supply that selector: in $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ the two individual conductor legs $(-3, -11)$ are *both* imaginary-quadratic (complex-type), and the unique *real* (orthogonal/Pfaffian) subfield is $\mathbb{Q}(\sqrt{33})$ — conductor **33**, the *product*, not the -3 leg. So a chiral Pfaffian would attach its $\frac{1}{2}$ to the conductor-33 combination, not to $\ln 3$ alone; and the register, being eleven real $\mathbb{Z}_2$ -seams, would if anything be the halved one (assignment ‘register-halved’, wildly wrong). **Verdict (pre-registered pass/fail): the $\frac{1}{2} \cdot \ln 3$ is fitting, not forced.** $\text{Pf}^2 = \det$ is a real $\frac{1}{2}$ *mechanism*, but its *assignment* to $\ln 3$ is underdetermined by internally-computable structure. The internal spectral route to S1 is therefore **RESOLVED** closed-negative — not ‘open pending computation’.

(verify_zs_a28_v1_2.py, block P.)

The disciplined exit. This is exactly the outcome ZS-A27 §7 anticipates: the **A–Q–Only No–Go** forbids closing the absolute scale from dimensionless data, and the spectral invariants — being natural-number combinations — only *bracket* the observed value ($10 \ln 2 + \ln 3 \rightarrow 3.9\times$ over, $12 \ln 2 \rightarrow 0.34\times$ under) without an internally-forced selector. The remaining legitimate channel uses the two *measured* anchors and so satisfies the No–Go: $\rho_\Lambda \sim c \Lambda_{\text{QCD}}^3 H_0 \approx 6 \text{ meV}$. The sealed value 8.190 was used by no computation.

§7.5. The hierarchy-exponent unification: one number, two costumes (v1.3)

v1.3 sharpens this. The audit recommended replacing the Coleman–Weinberg/Pfaffian engine with an $\text{SU}(2)_Z$ dimensional-transmutation engine. Carrying that out — and committing to a *single channel* (the gauge Pontryagin $\Omega_Z = (32\pi^2)^{-1} \text{tr } F_Z \tilde{F}_Z$, with the gravitational Pontryagin demoted to Appendix C, since the two have distinct geometric origins and no proof of forming an $\text{SU}(2)_Z$ doublet) — reveals that *both* engines compute the *same* quantity. The absolute-scale problem is one number, the hierarchy exponent

$$N \equiv \ln(\bar{\mathbf{M}}_p / \rho_\Lambda^{1/4}) = 69.16, \quad (\text{A28.41})$$

and the two engines are two parametrizations of it:

$$\text{Engine B (CW/Pfaffian): } N = 2\gamma_{\text{CW}} C_{\text{odd}}^{\text{sp}}; \quad \text{Engine A (SU}(2)_Z \text{ transmutation): } N = 16\pi^2/(b_0 g_Z^2). \quad (\text{A28.42})$$

The decisive observation (verify_zs_a28_v1_3.py, block N). At their *forced* values, *both engines miss*: Engine B with the un-halved $C_{\text{odd}}^{\text{sp}} = 8.723$ gives $N = 73.7$ (ρ 90 $\times$ short); Engine A with the normalized coupling $g_Z^2 = 0.258$ gives $N = 83.4$ ($\rho \sim 10^6$ short). Each matches observation only via an *unforced* tweak — and that tweak is, in each case, **exactly the sealed target**: Engine B requires $C_{\text{odd}}^{\text{sp}} = 8.190$ (the $\frac{1}{2} \cdot \ln 3$ value), Engine A requires $g_Z^2 = 0.3113$ (the prior sealed kill-switch). The identity $2\gamma_{\text{CW}} \cdot 8.190 = 16\pi^2/(b_0 \cdot 0.3113) = 69.16$ makes the equivalence exact.

Consequence. The engine swap does *not* close S1 — it relabels the same gap. The unforced quantity is the single exponent N , which is precisely $\ln(\bar{\mathbf{M}}_p/\rho_\Lambda^{1/4})$, the cosmological-constant hierarchy itself; the A–Q–Only No–Go forbids deriving it from dimensionless data, and we have now shown that *neither* internal engine evades this — they package N as a Pfaffian- $\frac{1}{2}$ or as a coupling, but do not predict its

value. This is the sharpest honest statement of the B3-scale obstruction the corpus has reached: *not* an impossibility theorem, but a demonstration that every internal route reduces to one undetermined number.

Cross-version (v1.3, ZS-A22 dependency). The pure-SU(2)_Z value $b_0 = 22/3$ differs from the ZS-A22 polyhedron rule $b_0 \in \{4/3, 5/3\}$ for the Z-sector mediator. These are *different objects* — A22's two-dimensional mediator versus the Z-spin Kramers gauge symmetry of §5 — so this is not a strict conflict; and in any case *all* three values give a confinement scale far from 2.3 TeV ($22/3 \rightarrow \sim \text{GeV}$; $4/3, 5/3 \rightarrow \text{sub-eV}$), confirming that no Z-sector confinement, on any reading, sets the dark-energy scale. A full version-dependency reconciliation is noted as remaining work.

The data channel is not free either (v1.3). The exit $\rho_\Lambda \sim c \Lambda_{\text{QCD}}^3 H_0$ must be handled with care: inputting the measured H_0 makes it a *calibration*, not a prediction, because H_0 is tied to ρ_Λ through Friedmann. A genuine prediction must *output* H_0 from $3\tilde{\mathbf{M}}_p^2 H^2 = \rho_m + \rho_r + c \Lambda_{\text{QCD}}^3 H$, which requires a *covariant H-linear action term*; the obvious candidate $\int d^4x \sqrt{-g} \nabla_\mu u^\mu$ is a boundary term and contributes nothing to the equations of motion. Constructing such a term is unbuilt work, so the absolute scale is, at present, neither internally derivable nor cleanly predicted from data — while the *relative* prediction $\Delta N_{\text{eff}} = 0.160$ (needing no absolute scale) remains falsifiable against CMB-S4.

§8. The gate ledger after v1.1, and what remains

Table 8.1. The seven B3 gates: v1.0 claim vs v1.3 audited status.

Gate	v1.0 claim	v1.3 status	Basis
B3-1	DERIVED-COND	PARTIAL	explicit variation pending (§3)
B3-2	DERIVED	DERIVED-COND	susceptibility Hessian PSD; numbers pending (§4)
B3-3	DERIVED-COND	PARTIAL	equilibrium q-boundary path (§6)
B3-4	OPEN	PARTIAL	SU(2) _Z + $g_Z = g_2$ forced; bundle/scale pending (§5)
B3-5	HYP→COND	RETIRED int.	scale = one exponent N ; no engine forces it (§7.5)
B3-6	DERIVED	OPEN	needs Dai–Freed η ; bosonic simplifies (§6)
B3-7	OPEN	OPEN	blind cosmology; H_0 must be output, not input (§7.5)

The net change across versions is honest in both directions. **Up:** the hidden gauge group is forced to **SU(2)_Z** and its coupling is computed ($g_Z = g_2$); B3-2 is PSD *by construction*; and — the v1.3 result — the absolute-scale problem is reduced to a *single number*, the hierarchy exponent $N = 69.16$. **Down:** B3-1, B3-3, and now B3-4 are corrected to **PARTIAL**, B3-6 to **OPEN**, the B3-5 torsion mis-label is **RETRACTED**, and the B3-5 internal spectral route is **RETIRED** (no engine forces N). The trunk is *not* closed — v1.0 over-claimed it — but it is rebuilt on physical objects, the gauge sector is forced, and the scale obstruction is now stated at its sharpest: every internal route reduces to one undetermined number, and even the data channel needs an unbuilt covariant term. **B3 remains OPEN**, with the remaining work (explicit variation; lattice susceptibility; a covariant H-linear term or a relative ΔN_{eff} test) clearly bounded. **Anti-numerology ledger.** The passing results are structural invariants — a mass dimension, a Lie-algebra dimension ($3 = \text{su}(2)$), a covariance-matrix positivity, a one-loop coefficient ($22/3$), and a Pfaffian identity ($\text{Pf}^2 = \det$) — none tunable, none using the sealed targets. The only dimensionful input is $\tilde{\mathbf{M}}_p$; $\gamma_{\text{CW}} = 38/9$

is anchored independently by the electroweak vev (the *even* invariant), not fitted to ρ_Λ . (**A, Q, dim Z**) = (35/437, 11, 2) **LOCKED**.

§9. Conclusion of Part I: why microscopic scale-generation fails

v1.1 is a correction and a consolidation. It withdraws three over-reaches and one error from v1.0 — the de Rham torsion was mis-labelled non-unitary (the conductor characters are unitary quadratic characters), the action-level selective-sequestering theorem was a component list with a coded identity, the rank-2 kernel was a Gram ansatz, and the membrane branch and parity-anomaly were declarations. In their place it proves two results that *are* rigorous. The hidden gauge group is **forced to be $SU(2)_Z$** by the spin- $\frac{1}{2}$ Kramers structure of the Z-sector, via a symplectic-automorphism theorem — tying the gauge group to the foundational Z-spin and supplying $b_0 = 22/3$ with no anomaly. The kernel and the scale coefficient are the *same* topological susceptibility, positive-semidefinite by construction, so B3-2 and B3-5 are one quantity. And the whole terminal scale reduces to a single number $C_{\text{odd}}^{\text{sp}}$, with a chiral **Pfaffian** $\frac{1}{2}$ the legitimate (non-numerological) structure that would give the pre-registered target $8.174 \rightarrow \rho_\Lambda^{1/4}$ at $1.15\times$ observed.

What is not done is stated plainly: the explicit Omnia-Sequestra variation (B3-1), the lattice susceptibility χ_{ab} (B3-2), the q-crossover dynamics fixing the branch (B3-3), the bundle/kinetic-term derivation and the matching scale (B3-4), the Dai–Freed anomaly (B3-6), and a covariant H-linear term for the data channel (B3-7). **B3 is OPEN**, neither closed nor proven impossible — and v1.3 states the obstruction at its sharpest: the absolute scale is a single number, the hierarchy exponent $N = \ln(\bar{M}_p/\rho_\Lambda^{1/4}) = 69.16$, and *every* internal engine — Coleman–Weinberg/Pfaffian or $SU(2)_Z$ transmutation — merely re-expresses N without forcing it, so the engine swap the audit proposed relabels rather than closes the gap. This is the disciplined terminus ZS-A27 §7 predicts, now made concrete: the cosmological-constant hierarchy is one undetermined number that dimensionless invariants cannot fix. The corpus standard is maintained — every v1.0 over-claim is retracted or scoped, the one striking coincidence ($\frac{1}{2} \cdot \ln 3$, equivalently $g_Z^2 = 0.3113$) is reported as *fitting rather than forced*, the verification is fail-closed, and the one dimensionful input is $\bar{M}_p$. The honest path forward is not another internal engine but the explicit trunk computations plus a falsifiable relative prediction ($\Delta N_{\text{eff}} = 0.160$) and, if a covariant H-linear term can be built, a parameter-free H_0 output tested blind against DESI.

———— PART II: THE X–Y VACUUM BOUNDARY LAW ————

§10. The reframe: vacuum energy is a UV/IR boundary, not a microscopic constant

Part I converged on a single negative fact: every microscopic engine merely re-expresses the one number $N = \ln(\bar{M}_p/\rho_\Lambda^{1/4}) = 69.16$ without predicting it, because the A–Q-Only No-Go forbids generating a large dimensionless hierarchy from locked dimensionless data. The resolution is not a better engine but a *category correction*. Quantum field theory computes the vacuum as the zero-point energy summed to the UV cutoff — the maximum a microscopic cell can hold, $\sim \bar{M}_p^4$ — a Y-sector (micro) quantity. What is observed is the faint IR floor that an extended X-sector (macro) region can carry. The 120-order ‘discrepancy’ compares a Y-ceiling to an X-floor: it is a mismatch of layers, exactly the X / Y separation the corpus posits, not a tuning failure.

Made precise, this is a *UV/IR relation*. The decisive identity (verify_zs_a28_v1_4.py, block H) is

$$N = \ln(\bar{\mathbf{M}}_p/\rho_\Lambda^{1/4}) = \frac{1}{2} \ln(\bar{\mathbf{M}}_p/H_0) - \frac{1}{4} \ln(3c^2), \quad (\text{A28.43})$$

so $N = 69.16$ is simply *half the logarithmic distance between the Planck and Hubble scales* (up to an $O(1)$ term). Equivalently

$$\rho_\Lambda^{1/4} = (3c^2)^{1/4} \sqrt[4]{(\bar{\mathbf{M}}_p H_0)} \quad \text{— the geometric mean of the UV ceiling and the IR floor.} \quad (\text{A28.44})$$

This **dissolves** the hierarchy problem: N is not a number a microscopic mechanism must manufacture; it is a ratio of two physical scales, which dimensionless invariants are not even asked to fix. The A–Q–Only No-Go forbids deriving an *absolute* scale from **A**, **Q** — it does *not* forbid the *ratio* of two given scales. The whole fine-tuning collapses from ~ 120 orders to a single **O(1)** number, the holographic coefficient $c^2 \approx 0.686$ — and an $O(1)$ dimensionless number is exactly the kind of object **A** = 35/437 and **Q** = 11 can hope to produce.

Honest framing. Equation (A28.43) is, on its own, the Friedmann definition of Ω_Λ rewritten; its *content* is the claim that ρ_Λ should be *read* as a boundary-saturation quantity ($\rho_\Lambda = c^2 \cdot 3\bar{\mathbf{M}}_p^2 H^2$) rather than as a fundamental constant. That reading is established external physics (next section); the Z-Spin task is then narrowed to deriving the one number c^2 and the cutoff L . The danger to guard against is *import-and-relabel*: holographic dark energy already exists, so the corpus earns its keep only if c^2 and L follow from **A**, **Q** rather than being dressed in Z-Spin language.

§11. The X–Y vacuum boundary law and its external lift

Let an X-sector region have radius L . The Cohen–Kaplan–Nelson bound [15] — an effective theory of size L with a UV cutoff cannot hold more energy than a black hole of that size without collapsing — gives $\rho L^3 \lesssim \bar{\mathbf{M}}_p^2 L$, i.e. $\rho_\Lambda \lesssim \bar{\mathbf{M}}_p^2 / L^2$. Saturating the bound at an IR horizon (Li normalization [16]),

$$\rho_\Lambda = 3 c^2 \bar{\mathbf{M}}_p^2 / L^2. \quad (\text{A28.45})$$

In the X / Y / Z reading this *is* the user’s boundary intuition: the X-sector vacuum (the ‘minimum nonzero spatial energy’) equals, across the Z-sector, the maximum energy a Y-sector horizon point can carry (its Schwarzschild capacity $M_{\text{BH}}(L) = 4\pi\bar{\mathbf{M}}_p^2 L$), with the Z-sector setting the saturation ratio. With L the de Sitter / Hubble horizon $1/H$, and Friedmann $3\bar{\mathbf{M}}_p^2 H^2 = \rho_{\text{tot}}$, equation (A28.45) gives $c^2 = \rho_\Lambda / \rho_{\text{crit}} = \Omega_\Lambda$.

External lift (frame H, v1.7 — static de Sitter entropy, not dynamical HDE). The frame rests on two established, non-Z-Spin results. The CKN UV/IR bound [15] supplies the *motivation* (a region of size L cannot exceed its black-hole energy), and the **Gibbons–Hawking / Wald** de Sitter entropy $S_{\text{ds}} = A_{\text{ds}}/4G = 8\pi^2\bar{\mathbf{M}}_p^2/H_\infty^2$ [18], which the corpus already carries (ZS-A23 §8, ZS-U1 §9.1), supplies the *mechanism*: combined with Friedmann $3\bar{\mathbf{M}}_p^2 H^2 = \rho_{\text{tot}}$, it fixes $\rho_\Lambda = 3\bar{\mathbf{M}}_p^2 H_\infty^2$ at the attractor and $\rho_\Lambda / \rho_{\text{crit}} = \Omega_{\Lambda,0} = 83/121$ today. **v1.7 NOTE:** this paper does *not* use Li’s dynamical future-event-horizon HDE [16] or the Hsu Hubble-cutoff $w = 0$ variant [17]; those were a v1.4/v1.5 detour, dropped in §14 because the corpus dark energy is constant Λ . CKN [15] and Li [16] are retained in the references only as the UV/IR *motivation* for a holographic bound, not as the dynamics. So $\rho_\Lambda \sim \bar{\mathbf{M}}_p^2 H^2$ is **DERIVED** (real physics, static de Sitter entropy); what remains Z-Spin-specific is the value $\Omega_{\Lambda,0} = 83/121$.

§12. The coefficient: $c^2 = \Omega_\Lambda = 83/121$ from corpus combinatorial holography

The holographic coefficient is not free: $c^2 = \Omega_\Lambda$, and the corpus already derives Ω_Λ . ZS-F2 §11.4 counts the late-time energy budget as a face-count saturation of the holographic bound on the 121-slot register: of $Q^2 = 121$ boundary slots, 6 + 32 are matter and **83** remain for the vacuum, giving

$$c^2 = \Omega_\Lambda = 83/121 = 0.6860, \quad c = \sqrt{83/121} = 0.8282. \quad (\text{A28.46})$$

This is a *face count*, not a tunable parameter, and it carries no back-solved numbers — none of the Part-I sealed values (8.190, 0.3113, 2.3 TeV) appears. The full budget and its checks:

Table 12.1. The (6, 32, 83)/121 holographic budget vs observation (verify_zs_a28_v1_6.py, block B83).

Quantity	Corpus value	Observed	Status
Ω_b	$6/121 = 0.0496$	≈ 0.0493	DERIVED-COND
Ω_{cdm}	$32/121 = 0.2645$	≈ 0.265	DERIVED-COND
$\Omega_\Lambda = c^2$	$83/121 = 0.6860$	≈ 0.685	DERIVED-COND
ρ_Λ ratio	$3c^2 \bar{M} P^2 H_0^2$	pred/obs = 1.01	DERIVED-COND
Ω_Λ/Ω_m	$83/38 = 2.184$	$2e^A = 2.167$ (0.8%)	DERIVED-COND

Where 83 lives (v1.5, block R.1). The register $Q^2 = 121 = 11 \otimes 11$ decomposes as the tensor $(3, 2, 6) \otimes (3, 2, 6)$ into nine blocks {XX 9, XZ 6, XY 18, ZX 6, ZZ 4, ZY 12, YX 18, YZ 12, YY 36}, summing to 121. The baryon slot **6** is the XZ block = $F(\text{cube})$; the cold slot **32** = $F(\text{truncated icosahedron}) = F(\text{icosahedron } 20) + F(\text{dodecahedron } 12)$, a Truncation–Dual identity (ZS-F2 §11.2, **PROVEN**; the truncated icosahedron is the unique mediator among the I_h Archimedean solids, §11.4); and $\Lambda = \mathbf{83}$ is the *remainder*, not a single block (the largest is YY = 36). The cold fraction was assigned *before* the cosmology check (§11.6.iv), and an anti-numerology Monte Carlo finds 0 of 500 000 random fractions beating 32/121.

The corpus derivation lineage (v1.6). 83 is not introduced by A28; it descends through four corpus papers. **ZS-F2 §11.4** derives the two independent integers by face counting: $\Omega_b = XZ/Q^2 = 6/121$ (the baryon is X-sector matter in the $H_X \otimes H_Z$ block, **DERIVED**) and $\Omega_{\text{cdm}} = F(\text{truncated icosahedron})/Q^2 = 32/121$, independently cross-checked by the Boundary Mode Theorem as $XQ - 1 = 32$. **ZS-U4** adopts the budget (6, 32, 83)/121 as a *locked late-time cosmological input* (not an observational coincidence).

ZS-A19 realises 32 as a conserved geometric Brown–Kuchař boundary-tension dust — an equivariant rank-32 projector in the 121-dimensional channel space (**DERIVED-COND**). **ZS-A20** derives $\rho_b/\rho_c = \text{Tr } P_b/\text{Tr } P_c = 6/32 = 3/16$ from a single-trace matrix current at the action level (**DERIVED** on global charge quantization). A separate route, ZS-A1/A5, gives $\Omega_\Lambda/\Omega_m = 2e^A = 2.167$ — a 0.8% *near-match* to the face-count $83/38 = 2.184$, *not* an identity (an unresolved cross-route, **OPEN**; 83 is not explained via $2e^A$).

The “82 + 1” re-expression — and why it is not extra forcing (v1.7). The complement can be re-bookkept in gauge-mode terms: the CDM slot count is $XQ = 33$; the Boundary Mode Theorem removes *one* Z_2 -odd gauge mode, leaving $r_c = XQ - 1 = 32$. Hence

$$r_\Lambda = Q^2 - r_b - r_c = Q^2 - XZ - (XQ - 1) = Q^2 - X(Z+Q) + \text{ind}_-(D_Z) = 121 - 39 + 1 = 82 + 1 = 83. \quad (\text{A28.45a})$$

This is *algebraically true*, and the +1 = $\text{ind}_-(D_Z)$ usefully *labels* the Z_2 -odd mode removed from CDM ($33 \rightarrow 32$). But **“82 + 1” adds no independent forcing**: it is the same complement $121 - 38$, re-grouped. The structural content is *entirely* in the forced integers 6 and 32; presenting 82 + 1 as if it were a separate derivation would be a mild numerology risk, so v1.7 marks it explicitly as a re-expression. It does, however, expose the *one* genuinely open denominator question: the corpus uses *case A* (the removed

mode stays a physical background mode, trace normalisation $Q^2 = 121$, $83 = 82 + 1$), but does not prove at action / physical-state-quotient level why it is not *case B* (a pure-gauge mode fully removed, denominator 120, $\Omega_{\text{vac}} = 82/120$). This is registered **OPEN**.

Honest status: two stages (v1.6). The status of 83 separates cleanly into a closed part and an open part:

Complement Theorem — $r_b = 6$, $r_c = 32 \Rightarrow r_\Lambda = 83$. **DERIVED-CONDITIONAL** (conditions: a common Q^2 -face algebra, orthogonal embedding, equal trace normalisation). The assignment *direction* is theorem-forced, not observation-selected: the baryon current J_B^μ is built from quark fields and is therefore an X-sector operator ($[J_B, O_Y] = 0$, $L_{XY} = 0$), so the baryon energy density must occupy the $X \otimes Z$ block = 6 (ZS-F5 Theorem B3.1); and the truncated icosahedron is the *unique* I_h Archimedean dual-face mediator (Theorem 11.4) $\rightarrow 32$. The swap (baryon = 32) failing by $\times 5.4$ is a cross-check, not the basis. **Honest asymmetry (v1.7):** the two legs are not equally strong — CDM = 32 is a self-contained polyhedral theorem (Truncation–Dual 20+12 + unique mediator), whereas baryon = 6 rests on the ZS-F5 B3.1 current-algebra argument, the softer of the two. The one residual is Φ_{face} 's equal weight $1/121$, narrowed to combinatorial (BV–BFV topological partition function, ZS-F0 §8.5) modulo a deferred categorical state-sum.

Vacuum Identification Hypothesis — $q_\Lambda \mapsto T_{\mu\nu}^\Lambda = -\rho_\Lambda g_{\mu\nu}$. **OPEN**. That the trace complement $q_\Lambda = 1 - q_b - q_c$ has $\tau(q_\Lambda) = 83/121$ is closed arithmetic; that this complement *is* the physical vacuum stress tensor is the unproven bridge.

In one line: **83 is the trace complement of the 38 matter boundary modes**, one unit of which is the Z_2 -odd mediation mode removed from CDM; the budget is derived, the vacuum-energy *identification* is the open bridge. **Crucially (v1.5 repair, §13)**, the identification $c^2 = \Omega_\Lambda = 83/121$ lives at the Q^2 -face level and is *not* the sector-weight Condition C — they are different objects, so c^2 does not depend on Condition C.

§13. The unification theorem (repaired): $B3 = P0 = [\text{face-budget } c^2]$, with Condition C decoupled

v1.4 stated a unification $B3 = P0 = C$ that conflated two corpus objects; v1.5 repairs it. The corrected theorem is cleaner.

Theorem A28.10 (Holographic reduction — conditional implication, DERIVED-CONDITIONAL on frame H). With $\rho_\Lambda = c^2 \cdot 3\bar{M}_p^2 H^2$: (i) **B3** (the value of ρ_Λ) is fixed once c^2 is known; (ii) at the Hubble/de-Sitter cutoff Friedmann gives $c^2 = \Omega_\Lambda$; (iii) $\Omega_\Lambda = 83/121$, the *face budget*. Hence the *conditional implication* [**Frame H** + Φ_{face} + a horizon-energy source coupling] $\Rightarrow$ **B3-relational** [$\rho_\Lambda = 3c^2\bar{M}_p^2 H^2$]. This is **not** an ‘if and only if’: $c^2 > 0$ by itself does *not* manufacture a positive cosmological source at the action level (Premise P0, ZS-A22, remains a separate debt — a constant $F = 1 + \mathbf{A}$ rescales $\bar{M}_p^2$ and sources *no* Λ). The face budget itself is DERIVED-CONDITIONAL (embedding Φ_{face} residual). **What changed, and why it is better.** v1.4's step (iv) wrote ‘ $\Omega_\Lambda = \text{Condition C} = (3,2,6)/11$ ’. That is a *conflation* (verify_zs_a28_v1_5.py, block R.1): $c^2 = 83/121 = 0.686$ is a Q^2 -face quantity, while the sector weight $(3,2,6)/11 = (0.273, 0.182, 0.545)$ has *no* component equal to 0.686 — Λ is not a sector slot at the Q level at all. So c^2 does *not* require Condition C; the two are **decoupled**. This *simplifies* the reframe: the vacuum coefficient now rests on the (largely proven) face budget, not on the harder operator-algebra dimension-match. Condition C becomes a *separate*, still-open sector-weight emergence problem that no longer gates B3.

The two weights are different objects (A19.NG1 resolved, block R.2). The corpus carries two weight triples that v1.4 risked conflating. **(3, 2, 6)/11** is the decomposition of the 11-dimensional *Hilbert register* ($X = 3$, $Z = 2$, $Y = 6$; sum = $Q = 11$) — kinematic sector sizes. **(3, 3, 5)/11** is the center trace of the 35-dimensional *operator algebra* $A_{Z5} = M_3 \oplus \mathbb{C} \oplus M_5$ (dim $9 + 1 + 25 = 35 =$ the numerator of $\mathbf{A}$) under its

natural unit embedding into M_{11} (multiplicities (1, 3, 1) solving $3n_3 + n_1 + 5n_5 = 11$). The corpus no-go A19.NG1 — that the natural embedding gives $(3,3,5)/11 \neq (3,2,6)/11$ — is therefore a *correct distinction* between a Hilbert register and an operator algebra, *not* a contradiction. The de Sitter maximum-entropy trace weights by whichever algebra it lives in; Condition C is the axiom asserting that the physical de Sitter sector decomposition matches the Hilbert register (3, 2, 6).

Condition C, now decoupled, is still sharpened but no longer B3's terminal. As in v1.4, CLPW [19] (de Sitter static-patch algebra is Type II₁, maximum-entropy state = unique trace) and Murray–von Neumann (trace = continuous dimension) close the *state-level* part, and the embedding non-uniqueness makes the residual *dimension-match* an irreducible emergence axiom. The difference in v1.5: this axiom concerns the *sector weights*, which the repair has shown are *not* what fixes c^2 . So Condition C is a real but *separate* open question; B3's coefficient does not wait on it.

Under the reframe the Part-I gates are reorganized: most were tied to the (failed) microscopic mechanism and are retired from the scale problem.

Table 13.1. Part-I gates under the Part-II reframe.

Part-I gate	Reframed status	Why
B3-1 vacuum cancellation	RETAINED	why local UV vacuum decouples from the source — still needed
B3-2 odd kernel	RETIRED	not the scale generator; superseded by holographic c^2
B3-3 flux branch	RETIRED	horizon state replaces the branch
B3-4 hidden gauge	RETIRED from B3	$SU(2)_Z$ may stand as separate physics
B3-5 transmutation scale	RETIRED	scale is a boundary quantity, not a condensate
B3-6 odd anomaly	DECOUPLED	separated from the scale problem
B3-7 cosmology	RETAINED	now the boundary-law falsification gate (§14)

The closure gates, updated in v1.6: **XB-1** the X–Y boundary theorem (derive $\rho_X V_X = c^2 E_{\max}^Y$ from a horizon Hamiltonian, **OPEN**); **XB-2** the coefficient $c^2 = 83/121$ — now *decoupled* from Condition C (§13) and **upgraded**: the assignment direction is theorem-forced (baryon = X-sector current $\rightarrow 6$; CDM = unique mediator $\rightarrow 32$, §12), so this is **DERIVED-COND** with only the equal-weight Φ_{face} state-sum residual; **XB-3** causal-scale uniqueness — **RETIRED** as inapplicable in v1.7: the corpus dark energy is constant Λ ($w = -1$, ZS-U1/U4) and the holographic scaling is *static* (de Sitter horizon entropy, A26/A27), where the Hubble, apparent, and event horizons coincide — so there is no horizon *choice* to derive (adopting constant Λ removes the dynamical-cutoff problem; it does not select a horizon); the dynamical- $w(z)$ detour is dropped (§14); **XB-4** dynamics $w(z)$ — now $w = -1$ (constant Λ), testable against DESI's dynamical-DE preference; **XB-5** the predictions $\Delta N_{\text{eff}}^{\text{BBN}} = 0.160$ (D/H, **DERIVED**) and $r = 0.0089$ (LiteBIRD) — the clean falsifiable channels (the *always-on* $\Delta N_{\text{eff}}^{\text{CMB}} = 0.160$ is FALSIFIED, $\Delta\chi^2 = +408$, §14). Of these, XB-2 is the corpus's genuine contribution; XB-5 is where the frame meets new data on parameter-free numbers.

§14. The dynamics: constant Λ , the de Sitter entropy, and the clean predictions

The corpus dark energy is constant Λ ($w = -1$), and that resolves the cutoff. Earlier drafts (v1.4/v1.5) put the coefficient $c = \sqrt{\Omega_\Lambda}$ into Li's dynamical future-event-horizon equation, yielding a running $w(z) \neq -1$. That was a *detour*: the corpus's actual dark energy is the de Sitter attractor with $w = -1$ *exactly* (ZS-U1 §9, ZS-U4), so ρ_Λ is constant, not $\rho_\Lambda(L(t))$. The holographic scaling $\rho_\Lambda \propto \bar{M}_p^2 H^2$ then arises *statically* from the de Sitter horizon entropy (the A26/A27 dilation-anomaly capstone): with $S_{\text{ds}} = 8\pi^2 \bar{M}_p^2 / H^2$,

the present-day relation and the asymptotic de Sitter relation are **DISTINCT** and must not be conflated:

$$\rho_\Lambda = 3 \Omega_{\Lambda,0} \bar{M}_p^2 H_0^2, \quad \Omega_{\Lambda,0} = 83/121 \quad (\text{TODAY, matter present}), \quad (\text{A28.48a})$$

$$\rho_\Lambda = 3 \bar{M}_p^2 H_\infty^2 = 24\pi^2 \bar{M}_p^4 / S_{\text{ds},\infty}, \quad H_\infty = \sqrt{(83/121)} H_0 \quad (\text{ASYMPTOTIC de Sitter}). \quad (\text{A28.48b})$$

Because ρ_Λ is constant, $\Omega_\Lambda(t) = \rho_\Lambda / (3\bar{M}_p^2 H^2)$ *varies*: today it is 83/121, and at the future attractor (matter diluted away) it tends to **1**. So **83/121 is today's energy fraction $\Omega_{\Lambda,0}$, while the de Sitter horizon-saturation coefficient is 1 — they are different objects**. The de Sitter *entropy* in (A28.48b) is evaluated at $H_\infty = \sqrt{(83/121)} H_0$, not at H_0 ; the capstone consistency $\ln(\bar{M}_p^4 / \rho_\Lambda) = 4N = 276.65$ (A26/A27) holds on the observed ρ_Λ .

At the de Sitter attractor the Hubble, apparent, and event horizons *coincide* (all equal $1/H_\infty$; in exact de Sitter $R_{\text{event}} = a \int dt/a = 1/H_\infty = R_{\text{Hubble}}$). So there is **no Hubble-vs-event-horizon choice to make** — the cutoff ambiguity (XB-3) arose *only* from forcing ρ_Λ to be dynamical ($\rho_\Lambda \propto 1/L(t)$). Adopting constant Λ does *not* derive which horizon; it **removes the dynamical-cutoff problem**. The honest status of XB-3 is therefore **RETIRED** — *retired-as-inapplicable*, not closed by derivation — and dropping the (now inapplicable) Li dynamics *resolves* the version conflict with ZS-U4 that v1.5 had only flagged.

B3-absolute — predicting H_∞ or H_0 from **A, Q** — remains a No-Go.

What the dynamics now predict, and the DESI test. The honest dynamical prediction is $w = -1$ (constant Λ , Λ CDM-like). **DESI DR2 (2025)** prefers *time-varying* dark energy ($w_0 > -1$, $w_a < 0$) over constant Λ at $2.8\text{--}4.2\sigma$ in its three-probe combination, crossing $w = -1$ near $z \approx 0.5$ — though the signal is parametrization-dependent and debated [22]. So *if* the DESI dynamical-DE preference hardens, the corpus's $w = -1$ is challenged — a clean, falsifiable statement. (This replaces v1.5's framing, in which the *vanilla holographic* $w(z)$ ran the *wrong* direction; with the detour dropped, the corpus simply predicts $w = -1$.)

The clean, scale-independent predictions. Independent of the cutoff and of any absolute scale, the corpus carries parameter-free numbers that meet *new* data directly. The first resolves a standing tension:

$$\Delta N_{\text{eff}}^{\text{BBN}} = \dim(Z) \cdot \mathbf{A} = 2\mathbf{A} = 70/437 = 0.1602, \quad (\text{A28.49})$$

a Z-sector dark-radiation contribution (ZS-U8/T1) that moves D/H from -2.3σ to **-0.05σ** . **Important correction (v1.6).** v1.5 over-stated this as an ‘always-on, 5σ CMB-S4’ prediction. The corpus's own Cobaya analysis (Step 2) shows the *always-on* scenario — $\Delta N_{\text{eff}}^{\text{CMB}} = 0.160$ present throughout recombination — is **FALSIFIED**: $\Delta\chi^2_{\text{CMB}} = +408.27$. What *survives* is the BBN-only activation (and gradual-decay) history. CMB-S4's role is therefore *not* a 5σ detection of 0.160; it is to **distinguish $\Delta N_{\text{eff}}^{\text{BBN}}$ from $\Delta N_{\text{eff}}^{\text{CMB}}$** — the activation/decay history — which is the genuinely informative measurement. The clean *near-term* CMB prediction, by contrast, is the tensor-to-scalar ratio

$$r = 0.0089 \quad (2.7\times \text{Starobinsky; high-significance at LiteBIRD nominal sensitivity } \delta r < 0.001), \quad (\text{A28.50})$$

together with $\eta_B = (6/11)^{35} = 6.12 \times 10^{-10}$ (0.02σ) and the proton-decay timescale $\tau_p = t_p \exp(5\pi/\mathbf{A}) \approx 2.56 \times 10^{34}$ yr (Hyper-K). These are the places the programme meets the world on parameter-free numbers.

Caveat (v1.7): LiteBIRD's mission target is a total tensor uncertainty $\delta r < 0.001$, so $r = 0.0089$ is a strong-signal candidate, but an exact '6 σ ' would require a dedicated Z-Spin forecast (foreground, lensing, sky mask) — we state it as high-significance-testable, not as a guaranteed 6 σ .

§15. Conclusion: from a failed engine to a falsifiable boundary law

This paper has two movements. **Part I** audited the microscopic route and converged on a negative result: the gauge group is forced, the kernel is a positive topological susceptibility, but every scale engine merely re-expresses one undetermined number $N = 69.16$, which the A–Q-Only No-Go forbids deriving. **Part II** reframes the problem along the user's X / Y duality, and the reframe *dissolves* the hierarchy: N is half the Planck–Hubble log distance, $\rho_\Lambda^{1/4}$ is the geometric mean $\sqrt{(\bar{\mathbf{M}}_p H_0)}$, and the ~ 120 -order tuning collapses to one O(1) number $c^2 = \Omega_\Lambda = 83/121$, which the corpus supplies by a face count whose integer (32) is a polyhedral theorem. **v1.5 then repairs the one real error in v1.4:** that face budget is a Q^2 -level object, *not* the Q -level sector-weight Condition C — so c^2 is *decoupled* from Condition C, which becomes a separate open problem rather than B3's terminal. The reframe is thereby *simpler*, not weaker. **v1.6 then grounds the coefficient and corrects two over-statements:** 83 is shown *derived* through the corpus lineage $ZS-F2 \rightarrow U4 \rightarrow A19 \rightarrow A20$ (with the deeper identity $83 = 82 + 1 = Q^2 - X(Z+Q) + \text{ind.}(D_Z)$), the assignment direction is upgraded from observation-selected to *theorem-forced*, XB-3 is *retired-as-inapplicable* under the corpus's constant- Λ de Sitter entropy (resolving the U4 conflict), and the ΔN_{eff} claim is corrected — $\Delta N_{\text{eff}}^{\text{BBN}} = 0.160$ survives while the always-on CMB version is falsified. The honest balance, stated precisely. **B3-relational** — the relation $\rho_\Lambda = c^2 \cdot 3\bar{\mathbf{M}}_p^2 H^2$ with $c^2 = 83/121$ — is **DERIVED-CONDITIONAL** (frame from the CKN UV/IR motivation + Gibbons–Hawking/Wald de Sitter entropy; coefficient from the face budget, with residual assignment-direction and embedding freedom). **B3-absolute** — predicting ρ_Λ or H_0 from $\mathbf{A}$, $\mathbf{Q}$, $\bar{\mathbf{M}}_p$ alone — is a *non-claim*, forbidden by the No-Go; the hierarchy is reframed as a ratio, not eliminated, with H_0 an external cosmological-state input. The *dynamics* are now *simpler*, not encumbered: with the dynamical-holographic detour dropped, the cutoff-selection problem (XB-3) is *retired-as-inapplicable* — the corpus dark energy is constant Λ ($w = -1$) and the holographic scaling is static (de Sitter entropy), so the U4 conflict is *resolved*. The honest dynamical prediction is therefore $w = -1$, mildly tensioned by DESI's dynamical-DE preference if it hardens; and the *clean* falsifiable channels are $\Delta N_{\text{eff}}^{\text{BBN}} = 0.160$ (D/H, -0.05σ) and $r = 0.0089$ (LiteBIRD) — with the always-on $\Delta N_{\text{eff}}^{\text{CMB}} = 0.160$ honestly recorded as FALSIFIED ($\Delta\chi^2 = +408$). **The epistemic frame matters:** for a zero-parameter theory there is no 'validation' — the mathematics is right or wrong by computation (subject to independent adversarial error-checking against shared blind spots), and the parameter-free predictions are confirmed or falsified by data. The user's intuition was decisive: the vacuum energy is not a number to be derived but a *relation between sectors*. v1.5 places that relation on a corrected, simpler footing and names the one number — ΔN_{eff} — on which the programme now cleanly meets the sky. ($\mathbf{A}$, $\mathbf{Q}$, $\dim \mathbf{Z}$) = (35/437, 11, 2) **LOCKED**.

The precise B3 classification (v1.7). To state exactly what is and is not closed, B3 splits four ways:

- B3-A** — A–Q-only absolute generation, $(\mathbf{A}, \mathbf{Q}, \bar{\mathbf{M}}_p) \rightarrow H_0$. **CLOSED-NEGATIVE** (PROVEN No-Go): a dimensionful scale cannot come from dimensionless $\mathbf{A}$, $\mathbf{Q}$ plus a single anchor. This is a result, not an open question.
- B3-B** — absolute scale allowing one extra dimensionful cosmological-state datum, $(\mathbf{A}, \mathbf{Q}, \bar{\mathbf{M}}_p, \text{datum}) \rightarrow H_0$. **OPEN** (re-explorable should a new boundary/initial datum be supplied).

B3-C — the present-day relation $\rho_{\Lambda,0} = 3 (83/121) \bar{M}_p^2 H_0^2$. **DERIVED-CONDITIONAL** (holds; uses H_0 as input — a reconstruction of ρ_{Λ} , not a prediction of it).

B3-D — the Vacuum Identification bridge $q_{\Lambda} \mapsto T_{\mu\nu}^{\Lambda} = -\rho_{\Lambda} g_{\mu\nu}$. **OPEN** — the *final* load-bearing item (= Premise P0). It is the single decidable-but-undecided internal question whose resolution would close B3-relational.

So the original B3 — the absolute late-time scale of ZS-A27 — is **not closed**. What this paper does is **reframe** (the ~ 120 -order hierarchy becomes a ratio, $\rho_{\Lambda}^{1/4} = \sqrt[4]{(\bar{M}_p H_0)}$), **ground** (the coefficient 83/121 via the corpus lineage ZS-F2→U4→A19→A20, with the forced integers 6 and 32 and the complement 83), and **localize** the residue to four named OPEN items — Φ_{face} 's equal-weight state-sum, the 121-vs-120 denominator, the UV cancellation Ward identity (Part I B3-1, **PARTIAL**), and above all the Vacuum Identification B3-D. **As a capstone, A28 closes the question of what B3 can and cannot be:** the microscopic-engine route is closed-negative, the budget ratio is conditionally fixed, the absolute scale is a No-Go in the A–Q-only sense, and the one bridge that remains — why the 83-complement is the physical constant vacuum stress tensor at today's H_0 — is the precise, decidable target for any successor.

Acknowledgements and Code Availability

verify_zs_a28_v1_7.py (fail-closed, SHA256

6278e29145fc098d5b673c30972ce8a1f1a6b18d67cf6a5a13c87cf916e40c7d) proves all 94 checks with real asserts (any failure exits nonzero): Part I — the character-unitarity retraction, the symplectic $\mathfrak{su}(2)$ solution, the susceptibility-Hessian covariance form, $g_Z = g_2$, and the hierarchy-exponent unification; Part II — the geometric-mean identity and the holographic budget; and the **v1.5 repair (block R)** — the $Q^2 = 121$ tensor decomposition showing 83/121 is a face budget (not the sector weight Condition C), the (3,2,6) Hilbert-register vs (3,3,5) operator-algebra distinction, the conserved-dust CDM (no coupling) giving a DESI-disfavoured $w(z)$, and the ZS-U4 version flag; and the **v1.6 additions** — the 83 derivation lineage (ZS-F2→U4→A19→A20), the structural identity $83 = 82 + 1$ (a re-expression, not extra forcing — §12), the theorem-forced assignment direction, and the ΔN_{eff} correction ($\Delta N_{\text{eff}}^{\text{BBN}} = 0.160$ survives, always-on $\Delta N_{\text{eff}}^{\text{CMB}}$ FALSIFIED at $\Delta\chi^2 = +408$); and the **v1.7 consistency pass** — retiring the contradictory dynamical- $w(z)$ checks, distinguishing $\Omega_{\Lambda,0} = 83/121$ (today) from the de Sitter coefficient 1 ($H_{\infty} = \sqrt[4]{(83/121)} H_0$), retiring XB-3 as inapplicable, and the four-way B3 classification. **Methodological note.** Part of the reframe and several repairs originated in external audit of v1.0–v1.4, in a user reframing intuition, and in AI-assisted exploration. The right epistemic frame for a *zero-parameter* theory is not ‘external validation’ but two distinct judgements: the mathematics is decidable (right or wrong by computation, subject to *independent adversarial error-checking* because author and AI can share blind spots — the corpus has seen true-conclusion/wrong-proof cases before), and the parameter-free predictions are *falsified or confirmed* by data, not fitted. A passing suite confirms only internal consistency. The load-bearing open items are the face-budget embedding Φ_{face} 's equal-weight categorical state-sum, the 121-vs-120 denominator, the UV-cancellation Ward identity (Part I B3-1, **PARTIAL**), and — above all — the Vacuum-Identification bridge B3-D ($q_{\Lambda} \rightarrow T_{\mu\nu}^{\Lambda} = -\rho_{\Lambda} g_{\mu\nu}$, = Premise P0), the single decidable-but-undecided internal item that would close B3-relational; the data channels are $\Delta N_{\text{eff}}^{\text{BBN}}/\tau/\text{DESI}$ (XB-3 is retired-as-inapplicable, not load-bearing). **On the loop.** The framing and judgements in v1.0–v1.7 are AI-assisted analysis; re-verifying decidable mathematics is legitimate error-detection but does not by itself break that loop — and v1.7, a consistency/honesty pass, does not break it either. The genuine next breaks are (a) deriving the Vacuum-Identification bridge B3-D from the

action (decidable-but-undecided internal mathematics — the real closure point of B3-relational); (b) data — pre-registering $r = 0.0089$ against LiteBIRD, or the $\Delta N_{\text{eff}}^{\text{BBN}}$ and DESI $w = -1$ tests; or (c) review by someone outside the loop. This work used Anthropic Claude; the author assumes full responsibility.

Appendix A. The symplectic-selection computation (Theorem A28.7)

In the basis $h = I$, $C = J_0 K$ with $J_0 = [[0,1],[1,0]]$, a general anti-Hermitian generator is $X = [[ia, b], [-\bar{b}, ic]]$ (a, c real, b complex). Commutation with the antilinear C is $X J_0 = J_0 \bar{X}$, whose four complex components reduce to the single real condition $a = -c$. The surviving generators are the traceless anti-Hermitian matrices $[[ia, b], [-\bar{b}, -ia]] = \{i\sigma_z, i\sigma_y, i\sigma_x\}$, a 3-dimensional real Lie algebra $\cong su(2)$. The exponential is the compact $SU(2) = USp(2)$. The construction is identical to the statement that $\mathbb{C}^2$ with a quaternionic structure is the quaternions $\mathbb{H}$, and $U(1, \mathbb{H}) = Sp(1) = SU(2)$.

Appendix B. External-anchor map and cross-version safety

A28 v1.1 *uses* (never modifies) ZS-M27 ($\mathbb{C}^2$ odd cohomology), ZS-Q14 (spin- $1/2$, $\chi_Z = -1$, the Kramers C), ZS-S15 ($J_Z = \gamma^5$), ZS-F9 ($\rho_Z = 0$), ZS-F23 (the $\mathbb{Z}_2$ -seam $\ln 2$), and ZS-A24–A27 (the B3 chain). The arithmetic conductors (ZS-M22) now enter only in the §7 *interpretation* of $C_{\text{odd}}^{\text{sp}}$, not in any proof — a deliberate consequence of removing the LOCATOR Gram matrix. No upstream result is altered; z^* (ZS-M1) is untouched.

Result / gate	External PROVEN anchor	Role
$SU(2)_Z$ (B3-4, §5)	compact symplectic group $Sp(1)=SU(2)$; Kramers/Wigner $T^2=-1$	forces the odd gauge group from spin- $1/2$
susceptibility (B3-2/5, §4)	positivity of pure-gauge topological susceptibility (lattice)	rank ≥ 1 ; sets residual coefficient
R_0 (B3-1, §3)	Kaloper–Padilla / Omnia-Sequestra [3,4]; Klinkhamer–Volovik [5–7]	removes constants; EW-kick residual
Pfaffian $1/2$ (B3-5, §7)	chiral-determinant = Pfaffian ($Pf^2=\det$); Dai–Freed η for B3-6	legitimate $1/2$ from chiral grading, not non-unitarity

Appendix C. The gravitational Pontryagin channel (demoted in v1.3)

v1.0–v1.2 carried two odd operators, the gauge and gravitational Pontryagin densities $\Omega_1^- = (32\pi^2)^{-1} \text{tr } F \tilde{F}$ and $\Omega_2^- = (384\pi^2)^{-1} R \tilde{R}$, and treated them as an $SU(2)_Z$ doublet. v1.3 demotes Ω_2^- : the two have *distinct geometric origins* (a hidden gauge bundle versus the spacetime curvature), and there is no proof that they transform as a single $SU(2)_Z$ representation. Since B3-2 requires only rank ≥ 1 , the single gauge channel $\Omega_Z = \Omega_1^-$ suffices and makes the topological susceptibility directly computable on a pure- $SU(2)$ lattice, the anomaly analysis simpler, and the artificial-doublet question moot. The gravitational channel is retained here only as a possible second contribution to the susceptibility matrix χ_{ab} ; whether it is independent (raising the rank to 2) is a **TARGET-COMPUTATION** for the capstone, not part of the v1.3 main line.

References

- [1] N. Kaloper and A. Padilla, Sequestering the Standard Model Vacuum Energy, Phys. Rev. Lett. 112, 091304 (2014), arXiv:1309.6562.
- [2] N. Kaloper and A. Padilla, Vacuum Energy Sequestering: The Framework and Its Cosmological Consequences, Phys. Rev. D 90, 084023 (2014), arXiv:1406.0711.
- [3] N. Kaloper, A. Padilla, D. Stefanyszyn, and G. Zahariade, Manifestly Local Theory of Vacuum Energy Sequestering, Phys. Rev. Lett. 116, 051302 (2016), arXiv:1505.01492.
- [4] B. Coltman, Y. Li, and A. Padilla, Cosmological consequences of Omnia Sequestra, JCAP 06, 017 (2019), arXiv:1903.02829.
- [5] F. R. Klinkhamer and G. E. Volovik, Self-tuning vacuum variable and cosmological constant, Phys. Rev. D 77, 085015 (2008), arXiv:0711.3170.
- [6] F. R. Klinkhamer and G. E. Volovik, Vacuum energy density kicked by the electroweak crossover, Phys. Rev. D 80, 083001 (2009), arXiv:0905.1919.
- [7] F. R. Klinkhamer and G. E. Volovik, Brane realization of q-theory and the cosmological constant problem, JETP Lett. 103, 627 (2016), arXiv:1604.06060.
- [8] R. Bousso and J. Polchinski, Quantization of four-form fluxes and dynamical neutralization of the cosmological constant, JHEP 06, 006 (2000), arXiv:hep-th/0004134.
- [9] X. Dai and D. S. Freed, η -invariants and determinant lines, J. Math. Phys. 35, 5155 (1994), arXiv:hep-th/9405012.
- [10] E. Witten, An SU(2) anomaly, Phys. Lett. B 117, 324 (1982).
- [11] D. B. Ray and I. M. Singer, R-torsion and the Laplacian on Riemannian manifolds, Adv. Math. 7, 145 (1971).
- [12] S. Weinberg, The cosmological constant problem, Rev. Mod. Phys. 61, 1 (1989).
- [15] A. G. Cohen, D. B. Kaplan, and A. E. Nelson, Effective field theory, black holes, and the cosmological constant, Phys. Rev. Lett. 82, 4971 (1999), arXiv:hep-th/9803132.
- [16] M. Li, A model of holographic dark energy, Phys. Lett. B 603, 1 (2004), arXiv:hep-th/0403127.
- [17] S. D. H. Hsu, Entropy bounds and dark energy, Phys. Lett. B 594, 13 (2004), arXiv:hep-th/0403052.
- [18] G. W. Gibbons and S. W. Hawking, Cosmological event horizons, thermodynamics, and particle creation, Phys. Rev. D 15, 2738 (1977).
- [19] V. Chandrasekaran, R. Longo, G. Penington, and E. Witten, An algebra of observables for de Sitter space, JHEP 02, 082 (2023), arXiv:2206.10780.
- [20] R. Bousso, A covariant entropy conjecture, JHEP 07, 004 (1999), arXiv:hep-th/9905177.
- [21] DESI Collaboration, DESI DR2 results II: measurements of baryon acoustic oscillations and cosmological constraints, arXiv:2503.14738 (2025).
- [22] M. Cortès and A. R. Liddle, On the dark energy crossing the phantom divide, Mon. Not. R. Astron. Soc. Lett. 544, L121 (2025).
- [ZS-A27] K. Kang, ZS-A27 v2.3: The absolute vacuum-energy scale B3 — four-root map $\{C1', R0, M1, S1\}$, Z-Spin Cosmology (2026).
- [ZS-A28 v1.0] K. Kang, ZS-A28 v1.0: Selective sequestering of a Z_2 -odd four-form doublet, Z-Spin Cosmology (2026) [superseded by this version on the flagged points].
- [ZS-Q14/M27/S15/F9/F23] K. Kang, Spin- $\frac{1}{2}$ boundary ($\chi_{Z=-1}$); Kostant cubic Dirac ($\mathbb{C}^2$ odd cohomology); $\gamma^5=J$; $\rho_Z=0$; $\mathbb{Z}_2$ -seam $\ln 2$, Z-Spin Cosmology (2026).

Version History

v1.7 (June 2026): CONSISTENCY + HONESTY pass (no new physics; kept in A28). **Fixed a real inconsistency:** v1.6's suite still PASSED the Li dynamical-holographic $w(z)$ (old block H.4) AND the vanilla- $w(z)$ check (R.3) alongside the constant- Λ result (R.4) — i.e. it passed both $w \neq -1$ and $w \equiv -1$ as ‘the corpus prediction’. v1.7 RETIRES the Li integration and the vanilla- w /desi checks, so the suite is **one consistent constant- Λ cosmology**. **Distinguished** $c^2 = 83/121 = \Omega_{\Lambda,0}$ (TODAY's fraction) from the de Sitter saturation coefficient 1 (asymptotic), with $H_\infty = \sqrt{83/121} H_0$; the de Sitter entropy uses H_∞ , not H_0 . **XB-3 $\rightarrow$ RETIRED-AS-INAPPLICABLE** (adopting constant Λ removes the dynamical-cutoff problem; it does not derive a horizon). **DEMOTED ‘83 = 82 + 1’** to a re-expression (it adds no forcing — the forcing is entirely in 6 and 32, and $83 = 121 - 38$). Noted baryon = 6 (ZS-F5 B3.1) is the softer leg vs CDM = 32 (a polyhedral theorem). Softened $r = 0.0089$ to

'high-significance at LiteBIRD nominal sensitivity ($\delta r < 0.001$)'. Added the **four-way B3 classification** (B3-A A-Q-only absolute CLOSED-NEGATIVE / B3-B with extra datum OPEN / B3-C today's relation DERIVED-COND / B3-D Vacuum-Identification OPEN). Conclusion: the original B3 is **not closed** — it is reframed, grounded, and localized; B3-D is the final OPEN bridge. Verification 94/94 PASS (fail-closed).

v1.6 (June 2026): GROUNDED 83 in the corpus derivation and CORRECTED two over-statements (kept in A28; no new paper number). **83 is DERIVED**, not introduced here: ZS-F2 §11.4 face-counting ($\Omega_b = XZ/Q^2 = 6/121$, $\Omega_{\text{cdm}} = F(\text{truncated icosahedron})/Q^2 = 32/121$, cross-check $XQ-1 = 32$) $\rightarrow$ ZS-U4 (locked late-time budget) $\rightarrow$ ZS-A19 (rank-32 conserved dust) $\rightarrow$ ZS-A20 ($\rho_b/\rho_c = 6/32$ at action level), with the deeper identity **$83 = 82 + 1 = Q^2 - X(Z+Q) + \text{ind}_-(D_Z)$** . Upgraded the assignment direction from observation-selected to **theorem-forced** (baryon = X-sector current $\rightarrow 6$; CDM = unique I_h mediator $\rightarrow 32$), so $c^2 = 83/121$ is DERIVED-CONDITIONAL — not a hidden fit; split 83's status into a Complement Theorem (DERIVED-COND) and a Vacuum-Identification Hypothesis (OPEN). **XB-3 DISSOLVES**: the corpus dark energy is constant Λ ($w = -1$, ZS-U1/U4) and the holographic scaling is static (de Sitter horizon entropy, A26/A27), where the horizons coincide — no Hubble-vs-event-horizon choice; dropped the dynamical- $w(z)$ detour, which RESOLVED the ZS-U4 conflict v1.5 had flagged. Made Theorem A28.10 a *conditional implication*, not an iff. **Corrected** ΔN_{eff} : $\Delta N_{\text{eff}}^{\text{BBN}} = 0.160$ survives (D/H , -0.05σ) but the always-on $\Delta N_{\text{eff}}^{\text{CMB}} = 0.160$ scenario is FALSIFIED (corpus Cobaya Step 2, $\Delta\chi^2 = +408$); CMB-S4 distinguishes the activation history, and the clean near-term channel is $r = 0.0089$ (LiteBIRD). Verification 88/88 PASS (fail-closed).

v1.5 (June 2026): REPAIRED the v1.4 unification conflation (kept in A28; no new paper number). Theorem A28.10 *decouples* the coefficient from Condition C: $c^2 = \Omega_\Lambda = 83/121$ is a **$Q^2 = 121$ face budget** (the 9-block tensor $(3,2,6) \otimes (3,2,6)$; $\Lambda = 83$ the remainder, not a sector slot), *not* the **$Q = 11$ sector weight** $(3,2,6)/11$ — so c^2 no longer waits on the harder Condition C (a simplification). Resolved A19.NG1: $(3,2,6)/11$ (Hilbert register) and $(3,3,5)/11$ (operator algebra $A_{ZS} = M_3 \otimes \mathbb{C} \otimes M_5$, $\dim 35$) are distinct objects, not a contradiction. Honest $w(z)$: corpus CDM is conserved dust (A19, no coupling) $\Rightarrow$ vanilla holographic $w(z)$, DESI-disfavoured; a coupling needs an A19 modification (new structure, not derivation). Flagged: XB-3 cutoff **OPEN** (Hubble vs event-horizon), ZS-U4 constant- Λ conflict. Promoted $\Delta N_{\text{eff}} = 2A = 0.160$ (scale-independent, Cobaya step-1 PASS) to the primary falsifiable prediction. Clarified B3-relational (DERIVED-COND) vs B3-absolute (non-claim, No-Go). Verification 69/69 PASS (fail-closed).

v1.4 (June 2026): Added Part II — the X–Y holographic reframe. Hierarchy dissolves ($N = \frac{1}{2} \ln(\bar{M}_p/H_0)$), $c^2 = \Omega_\Lambda = 83/121$, B3 = P0 = C unified (*repaired in v1.5*), Li $w(z)$ vs DESI. Verification 55/55 PASS.

v1.3 (June 2026): Single channel + single engine; the hierarchy-exponent unification (both internal engines re-express $N = 69.16$); B3-4 coupling $g_Z = g_2$. 40/40 PASS.

v1.2 (June 2026): Chiral-Pfaffian: $\frac{1}{2}$ not forced onto $\ln 3$ (real subfield is conductor 33); spectral route RETIRED. 27/27 PASS.

v1.1 (June 2026): Corrected v1.0: RETRACTED non-unitary torsion; DEMOTED Thms A28.2/A28.4/A28.3/A28.5; NEW Theorems A28.7/A28.8/A28.9. 21/21 PASS.

v1.0 (June 2026): Initial release — the ZS-A27 §8.2 forward program. Superseded on the flagged points by v1.1–v1.5.